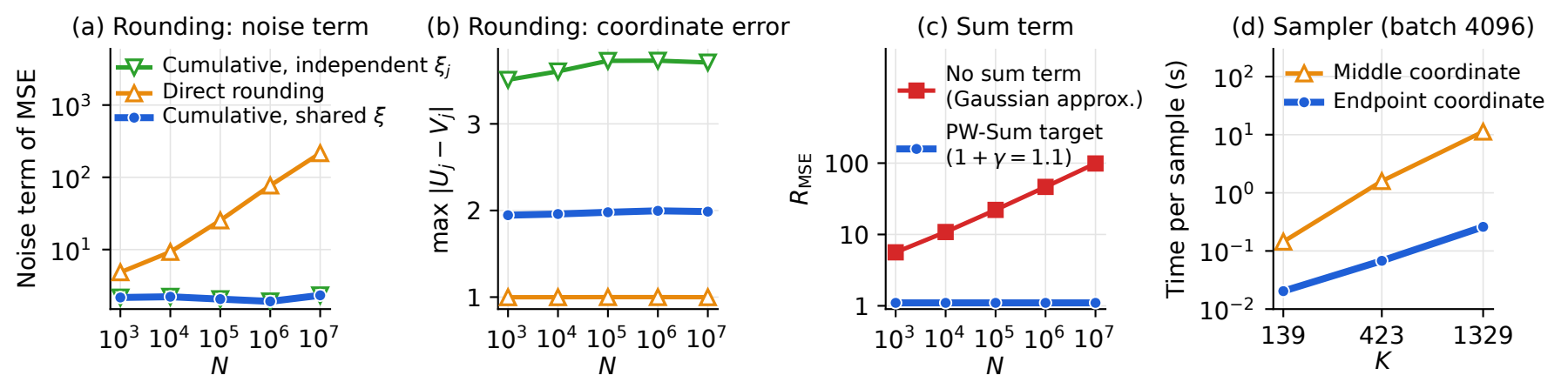
(a) Rounding: noise term
Noise term of MSE
Cumulative, independent ξ_j
Direct rounding
Cumulative, shared ξ
10^3
10^2
10^1
10^3 10^4 10^5 10^6 10^7
N
(b) Rounding: coordinate error
max $|U_j - V_j|$
3
2
1
10^3 10^4 10^5 10^6 10^7
N
(c) Sum term
R_{MSE}
No sum term
(Gaussian approx.)
PW-Sum target
$(1+\gamma = 1.1)$
100
10
1
10^3 10^4 10^5 10^6 10^7
N
(d) Sampler (batch 4096)
Time per sample (s)
Middle coordinate
Endpoint coordinate
10^2
10^1
10^0
10^{-1}
10^{-2}
139
423
1329
K